Reza Saadati Fard

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Summary

- Medical AI & Machine Learning Researcher
- Clinical Decision Support & Healthcare Analytics
- Computer Vision for Medical Images
- Generative AI, LLMs & Multimodal AI
- RL/RLHF, Fine-Tuning, RAG & Agentic AI
- GPU Computing, Distributed ML & MLOps

Technical Skills

- **Medical AI & LLMs:** Medical image analysis, clinical decision support, multimodal LLMs/VLMs, fine-tuning, RAG/visual RAG, semantic search, agentic reflection, multi-agent AI workflows
- **Machine Learning & Deep Learning:** Classification, regression, model evaluation, CNNs, Transformers, vision-language models, sequence/time-series modeling, reinforcement learning (RLHF)
- **Programming & Data:** Python, SQL, R, MATLAB, NumPy, Pandas, Scikit-learn, data preprocessing, feature engineering, structured/unstructured healthcare data
- **Scalable AI Systems:** GPU-based model training/inference, distributed computing concepts, AWS, Azure, Docker, Git/GitHub, Linux, CI/CD workflows, API-based ML systems

Professional Experience

Research Intern, Healthcare AI - Centific Aug 2026 – Present

- Research reinforcement learning methods, including RLHF-style fine-tuning, to improve the medical reasoning, safety, and reliability of LLMs and clinical foundation models.
- Design agentic AI workflows for multi-step medical reasoning, planning, and tool use that combine traditional ML and generative models to support clinical decision-making.
- Curate and preprocess HIPAA-compliant healthcare data, including EHRs, prescriptions, and medical images, into datasets for fine-tuning and evaluation, and run experiments aimed at publication.

Research Assistant - CareLab, Worcester Polytechnic Institute 2021 – Present

- Designed medical AI systems for pressure ulcer image classification, wound assessment, and clinical decision-support applications.
- Developed multimodal LLM pipelines using prompt design, fine-tuning, RAG/visual RAG, and agentic reflection to improve classification and rationale generation.
- Built reproducible and scalable ML workflows for medical image preprocessing, label handling, model evaluation, error analysis, and GPU/cloud-oriented experimentation using Python, Git, Linux, Docker, and AWS/Azure tools.

Deep Learning Engineer - Ortho BioMed, Canada 2019 – 2021

- Designed deep learning and statistical ML models for organ donor-recipient matching and healthcare analytics in a medical technology environment.
- Developed data preprocessing, feature extraction, model training, validation, and performance evaluation workflows for large medical datasets, including scalable and GPU-aware deep learning experiments.
- Contributed to AI-enabled decision-support tools and helped move ML prototypes toward production-ready healthcare software.

Full-Stack Software Engineer - Vesis Inc. 2012 – 2016

- Developed production backend services, APIs, database-driven applications, and reporting systems using C# and SQL Server.
- Built reliable data workflows and maintainable software systems supporting data processing and application deployment.

Education

Ph.D., Computer Science - Worcester Polytechnic Institute 2023 – Present
M.S., Computer Science - Worcester Polytechnic Institute 2021 – 2023
M.Sc., Electrical Engineering - Isfahan University of Technology 2015 – 2018
B.Sc., Electrical Engineering - Shahid Chamran University 2008 – 2012

Selected Achievements

- **3 U.S. Patents (2024–2025):** Wound Assessment and Classification Systems; Wound Image Gathering and Clarification Methods; Organ Donor Identification and Matching Systems.
- **National Health Grant, Canada:** Contributed to an award-winning ML system for improving organ transplantation outcomes.
- **WPI Graduate Research Innovation Exchange, People's Choice Award (2024).**
- **Publications:** 14 peer-reviewed publications in ML, AI, LLMs, medical AI, and multimodal learning. Full list: Google Scholar

Reference

- **Prof. Emmanuel Agu** — Professor of Computer Science, Worcester Polytechnic Institute (Ph.D. Advisor), emmanuel@wpi.edu